

One-to-one and Onto Transformation

Definition (One-to-one transformations). A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if, for every vector b in $\mathbb{R}^m$, the equation $T(x) = b$ has at most one solution x in $\mathbb{R}^n$.

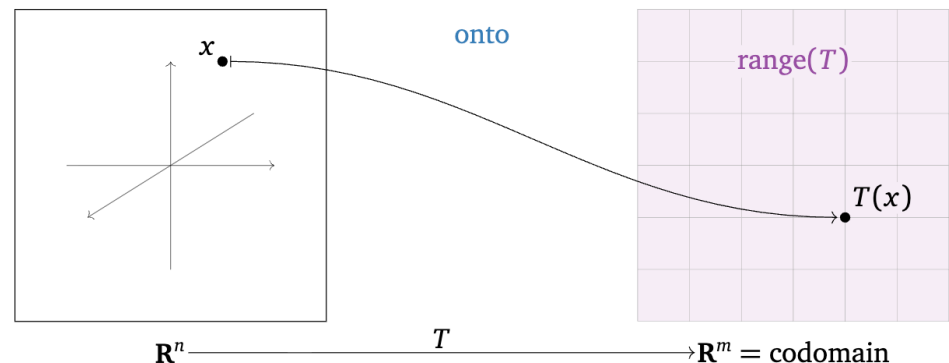
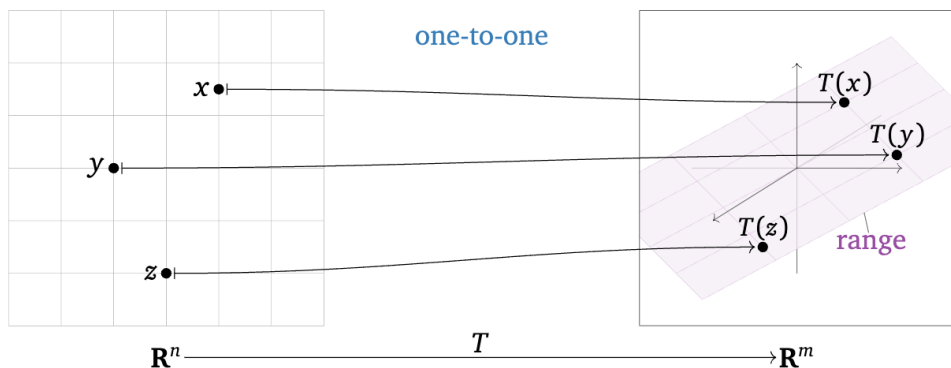
Theorem (One-to-one matrix transformations). Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation. The following statements are equivalent:

1. T is one-to-one.
2. For every b in $\mathbb{R}^m$, the equation $T(x) = b$ has at most one solution.
3. For every b in $\mathbb{R}^m$, the equation $Ax = b$ has a unique solution or is inconsistent.
4. $Ax = 0$ has only the trivial solution.
5. The columns of A are linearly independent.
6. A has a pivot in every column.
7. The range of T has dimension n .

Definition (Onto transformations). A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if, for every vector b in $\mathbb{R}^m$, the equation $T(x) = b$ has at least one solution x in $\mathbb{R}^n$.

Theorem (Onto matrix transformations). Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation. The following statements are equivalent:

1. T is onto.
2. $T(x) = b$ has at least one solution for every b in $\mathbb{R}^m$.
3. $Ax = b$ is consistent for every b in $\mathbb{R}^m$.
4. The columns of A span $\mathbb{R}^m$.
5. A has a pivot in every row.
6. The range of T has dimension m .



Linear Transformation

Definition. A *linear transformation* is a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfying

$$T(u + v) = T(u) + T(v)$$

$$T(cu) = cT(u)$$

for all vectors u, v in $\mathbb{R}^n$ and all scalars c .

Facts about linear transformations. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then:

1. $T(0) = 0$.
2. For any vectors $v_1, v_2, \dots, v_k$ in $\mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_k$, we have

$$T(c_1 v_1 + c_2 v_2 + \dots + c_k v_k) = c_1 T(v_1) + c_2 T(v_2) + \dots + c_k T(v_k).$$

Standard coordinate vectors.

The *standard coordinate vectors* in $\mathbb{R}^n$ are the n vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \dots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{pmatrix}, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

The i th entry of e_i is equal to 1, and the other entries are zero.

From now on, for the rest of the book, we will use the symbols $e_1, e_2, \dots$ to denote the standard coordinate vectors.

Multiplying a matrix by the standard coordinate vectors. If A is an $m \times n$ matrix with columns $v_1, v_2, \dots, v_m$, then $Ae_i = v_i$ for each $i = 1, 2, \dots, n$:

$$\begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix} e_i = v_i.$$

In other words, multiplying a matrix by e_i simply selects its i th column.

$$\text{If } A = [\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3 \quad \cdots \quad \vec{a}_n]$$

$$T(\vec{e}_1) = \vec{a}_1, \quad \dots, \quad T(\vec{e}_n) = \vec{a}_n$$

We saw in the above example that the matrix for counterclockwise rotation of the plane by an angle of θ is

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$